# Transcriptomic Analysis Reveals Distinct Responses in APP- and PSEN-Related Subtypes of Familial Alzheimer's Disease

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## INTRODUCTION

**Familial Alzheimer's Disease** (FAD) is a rare, inherited form of Alzheimer's caused by **mutations in the presenilin-1 (PSEN1), presenilin-2 (PSEN2), and amyloid-β precursor protein (APP) genes**, which alter amyloid-β (Aβ) production and contribute to Alzheimer's pathology. In this study, we focused on **APPV717I** and **PSEN1A79V** mutations. We performed **differential expression gene (DEG) analysis** and **gene set enrichment analysis (GSEA)** to identify gene expression patterns associated with these mutations, as well as a **network analysis** to explore connections between DEGs and their potential roles in disease mechanisms.

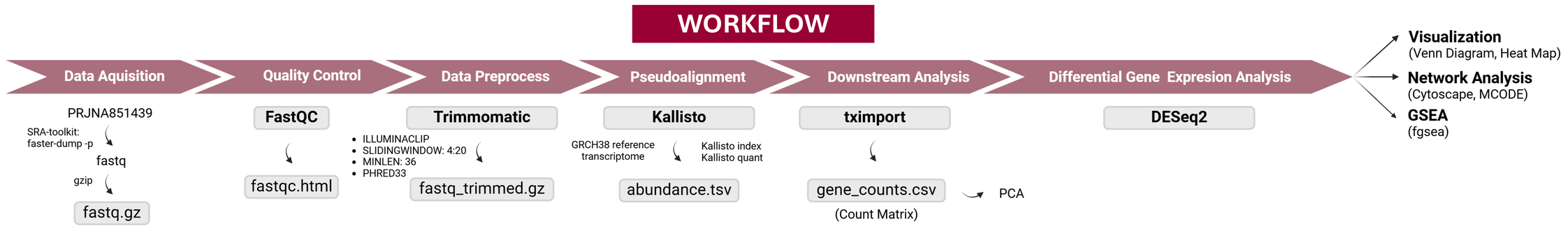
**Aim of the study**

Identify differences and similarities in gene expression patterns between APPV717I and PSEN1A79V mutations and uncover potential molecular pathways of Familial Alzheimer's Disease.

## DATA AVAILABILITY

Bulk RNA-seq data was purified from human WT, PSEN1_A79V and APP_V717I hiPSC-derived neurons (replicates, n=3). Paired end reads with length of 100bp were generated using Illumina HiSeq 4000. Raw reads (before adapter trimming) can be found in NCBI's SRA under accession number PRJNA851439.

## WORKFLOW




## RESULTS

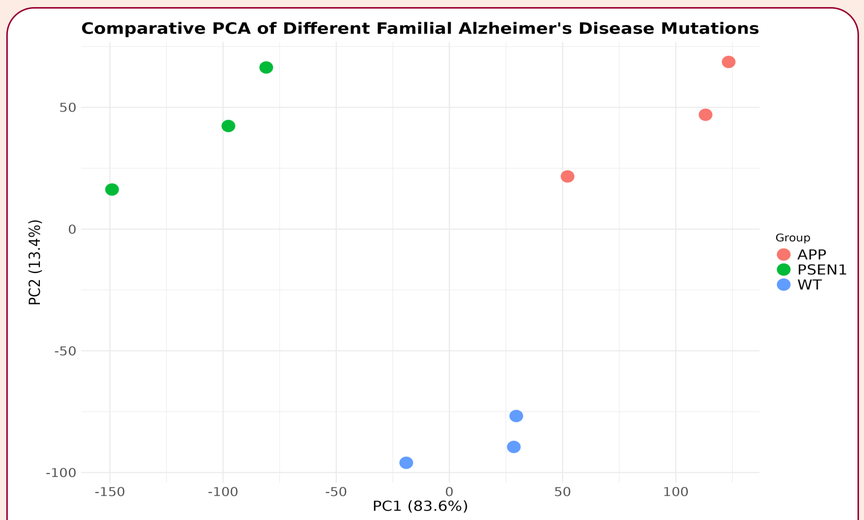



**Fig 1. PCA plot of samples.** The plot shows clear group separation, with PC1 (83.6% variance) and PC2 (13.4%). WT separates from AD samples, while APP and PSEN1 variants also differ along PC1, indicating distinct transcriptional profiles.

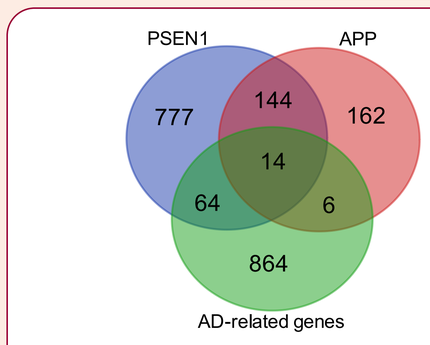



**Fig 2. Venn diagram showing the overlap between differentially expressed genes (DEGs) from APP and PSEN1 mutation samples and AD-related genes curated from the Agora AD Knowledge Portal. 14 genes** are shared across APP and PSEN1 mutations and are associated with known AD pathways, suggesting they play a central role in AD pathology. **144 genes** are shared between APP and PSEN1 DEGs but are not yet associated with AD, representing potential novel targets for further investigation into mutation-specific mechanisms or new AD pathways.

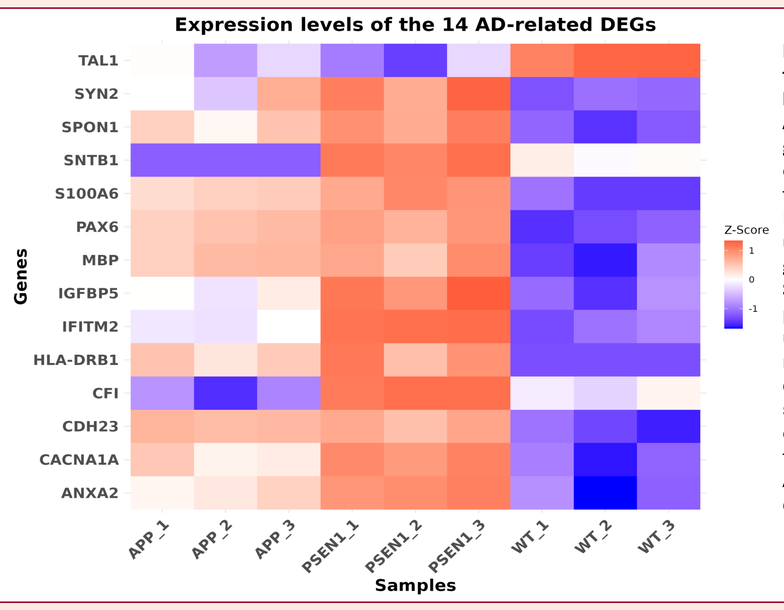



**Fig 3. Heatmap of Z-scores for DEGs that overlap between APP, PSEN1 and AD-related genes.** Most genes show opposing expression patterns between the mutations (APP and PSEN1) and the WT group, underscoring contrasting gene regulation pathways. SNTB1 and CFI show distinct patterns with significant upregulation in PSEN1-mutated samples but downregulation in APP samples compared to the control (WT), indicating that the two mutations may drive Alzheimer's Disease through different mechanisms.

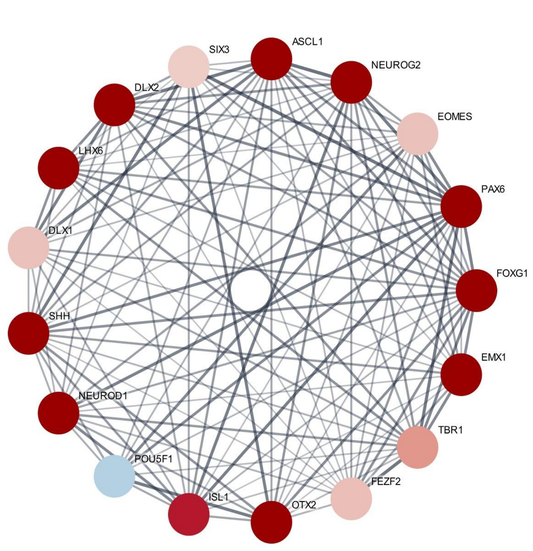



**Fig 4. Cluster identified within the network of 144 differentially expressed genes (DEGs) shared between two mutations, not previously associated with Alzheimer's Disease (AD).** Nodes represent individual genes, color-coded by $log_2FC$ (red for upregulated, blue for downregulated genes) in a continuous scale. Key functional terms, including neurogenesis and plasticity, highlight potential novel pathways and mechanisms involved in AD pathology. It is noteworthy that only POU5F1 appears to be downregulated in the cluster.

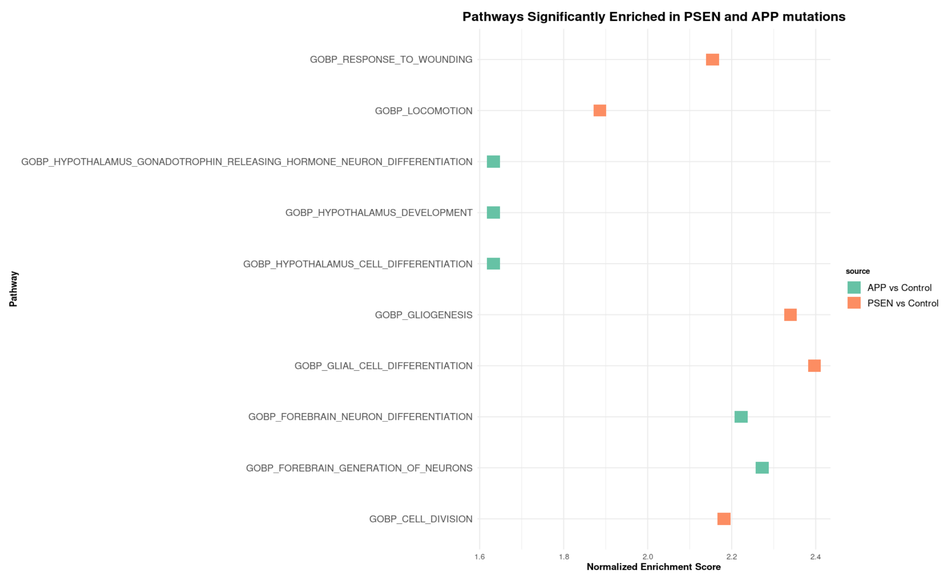



**Fig 5. Enriched pathways in PSEN and APP mutations (GSEA).** The plot displays the normalized enrichment scores (NES) for various biological pathways associated with PSEN and APP mutations compared to controls. Pathways are categorized by source: APP vs Control (green) and PSEN vs Control (orange). Enrichment scores reflect the magnitude of pathway dysregulation, highlighting differences in processes such as hypothalamus development, gliogenesis, and cell division. We find different enriched biological processes among the two subtypes of FAD, which could be linked to the pathological characteristics of each subtype.

Notably, APP mutations, usually involved in amyloid beta plaque production, show enrichment in pathways related to forebrain neuron differentiation, linking this to neurodegenerative processes in Alzheimer's disease; whereas PSEN1 mutations, key components of the gamma-secretase complex, exhibit stronger associations with pathways like cell division and gliogenesis, potentially reflecting a broader impact on neural homeostasis and cell signaling disruptions common in AD, as well as a greater link to motor dysfunction, generally more common in this subtype.

## CONCLUSIONS

- The **PCA plot** (Fig. 1) highlights **distinct patterns** between the two mutations analyzed in this study. **Differential expression analysis** further investigated these differences, identifying both common and unique gene expression responses, as visualized in the **Venn diagram** (Fig. 2).
- Among the **DEGs** associated with AD pathology, we identified two genes, **CFI** and **SNTB1**, that exhibit opposing expression patterns in the two mutations. Both genes are downregulated in APP-mutated individuals but upregulated in PSEN-mutated samples. **SNTB1** plays a critical role in synaptic organization and signal transduction, while **CFI** is a key serine protease in the c**omplement system**, with its dysregulation linked to impaired amyloid-beta clearance and chronic neuroinflammation.
- **Network analysis** revealed a cluster (Fig. 4) with important association to **neurogenesis** and **neuroplasticity**, as well as **signal transduction** and **synaptic plasticity**. In this cluster, dysregulation of elements like **POU5F1**, might have a potential impact in AD pathology.
- The **distinct pathway enrichments observed in APP and PSEN1** mutations (Fig. 5) highlight **subtype-specific mechanisms** in familial Alzheimer's disease, with APP mutations emphasizing amyloid-beta-related **neurodegeneration** and PSEN1 mutations linked to **disrupted neural homeostasis** and **motor dysfunction**.

## REFERENCES

1. Agora AD Knowledge Portal hhttps://agora.adknowledgeportal.org/
2. Valdes, P., Caldwell, A.B., Liu, Q. *et al*. Integrative multiomics reveals common endotypes across *PSEN1*, *PSEN2*, and *APP* mutations in familial Alzheimer's disease. *Alz Res Therapy* **17**, 5 (2025). https://doi.org/10.1186/s13195-024-01659-6
3. Salem, N. A., Mahnke, A. H., Tseng, A. M., Garcia, C. R., Jahromi, H. K., Geoffroy, C. G., & Miranda, R. C. (2021). A novel Oct4/Pou5f1-like non-coding RNA controls neural maturation and mediates developmental effects of ethanol. *Neurotoxicology and Teratology, 83,* 106943. https://doi.org/10.1016/j.ntt.2020.106943